

## Assignment Title:

### Agriculture equipment rental and farm resource management platform

#### 1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

#### Project Vision

To develop a digital platform that enables farmers to rent agricultural equipment, manage farm resources, and improve farming productivity by connecting equipment owners, farmers, and administrators through a simple and secure online system.

#### Project Objectives

- Provide online equipment rental services.
- Enable efficient farm resource management.
- Reduce equipment ownership costs for farmers.
- Increase equipment utilization for owners.
- Offer secure online booking and payment.
- Improve transparency through real-time tracking and notifications.

#### Key Stakeholders

| Stakeholder            | Role                                           |
|------------------------|------------------------------------------------|
| Farmers                | Rent equipment and manage farm resources       |
| Equipment Owners       | List equipment and receive rental income       |
| Administrator          | Manage users, equipment, bookings, and reports |
| Delivery Partner       | Deliver and collect rented equipment           |
| Payment Gateway        | Process secure online payments                 |
| Agricultural Suppliers | Provide fertilizers, seeds, pesticides, etc.   |

#### 2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:  
*As a <user>, I want <feature>, so that <benefit>.*

#### Epic 1: User Management

##### User Story 1

**As a farmer, I want to register an account so that I can rent farming equipment online.**

## User Story 2

**As an equipment owner, I want to create my profile so that I can list my equipment for rent.**

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## Epic 2: Equipment Management

### User Story 3

**As an equipment owner, I want to add equipment details so that farmers can view and rent them.**

### User Story 4

**As a farmer, I want to search available equipment so that I can rent suitable machinery.**

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## Epic 3: Booking Management

### User Story 5

**As a farmer, I want to book equipment for a selected date so that I can use it during my farming activities.**

### User Story 6

**As an equipment owner, I want to approve or reject bookings so that I can manage my equipment schedule.**

## Epic 4: Payment Management

### User Story 7

**As a farmer, I want to pay online so that my booking is confirmed instantly.**

## Epic 5: Resource Management

### User Story 8

**As a farmer, I want to record fertilizers, seeds, labor, and expenses so that I can monitor farm resources.**

## Epic 6: Reports

### User Story 9

**As an administrator, I want to generate reports so that I can monitor platform performance.**

#### 3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

### User Story 1

**Given** a new user visits the website

**When** valid registration details are entered

**Then** the account should be created successfully.

### User Story 3

**Given** the equipment owner is logged in

**When** equipment details are submitted

**Then** the equipment should appear in the available equipment list.

### User Story 5

**Given** equipment is available

**When** the farmer books the equipment

**Then** the booking should be confirmed and saved.

### User Story 7

**Given** booking is confirmed

**When** payment is completed successfully

**Then** payment status should change to "Paid".

## User Story 8

**Given** the farmer is logged in

**When** farm resources are entered

**Then** the system should save and display the updated records.

### 4. Prioritize the Product Backlog

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

| Priority | Product Backlog Item      | Business Value |
|----------|---------------------------|----------------|
| High     | User Registration & Login | Essential      |
| High     | Equipment Listing         | Essential      |
| High     | Equipment Search          | Essential      |
| High     | Equipment Booking         | Essential      |
| High     | Online Payment            | Essential      |
| Medium   | Resource Management       | High           |
| Medium   | Notifications             | Medium         |
| Medium   | Booking History           | Medium         |
| Low      | Ratings & Reviews         | Low            |
| Low      | Reports & Analytics       | Low            |

### 5. Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

## Sprint 1 User Stories

- User Registration
- User Login
- Equipment Listing
- Equipment Search
- Equipment Booking

## Development Tasks

### Registration Module

- Design registration page
- Validate user details
- Store user information

### Login Module

- Create login page
- Authenticate users
- Manage sessions

### Equipment Module

- Add equipment
- Edit/Delete equipment
- Display equipment list

### Search Module

- Search by equipment type
- Search by location
- Filter by availability

### Booking Module

- Book equipment
- Save booking
- Display booking status

## 6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

| User Story        | Story Points | Justification               |
|-------------------|--------------|-----------------------------|
| Registration      | 3            | Simple forms and validation |
| Login             | 2            | Authentication only         |
| Equipment Listing | 5            | CRUD operations             |

| User Story          | Story Points | Justification                |
|---------------------|--------------|------------------------------|
| Equipment Search    | 5            | Filtering and searching      |
| Equipment Booking   | 8            | Booking logic and validation |
| Payment Integration | 8            | API integration and security |
| Resource Management | 5            | Multiple data entries        |
| Reports             | 3            | Report generation            |

**Estimation Technique Used:** Fibonacci Sequence (2, 3, 5, 8)

#### 7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

##### Sprint Goal

Develop the core functionalities of the Agriculture Equipment Rental and Farm Resource Management Platform, enabling farmers to register, log in, search equipment, list equipment, and book available machinery.

## Evaluation Rubrics:

| Criteria                                                         | Excellent (5 Marks)                                                                                                                                 | Good (3 Marks)                                                                                       | Satisfactory (2 Marks)                                                                  | Needs Improvement (1 Mark)                                                                                   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>1. Project Vision &amp; Stakeholder Identification</b>        | Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified. | Vision is clear with minor omissions. Most stakeholders are correctly identified.                    | Vision is partially defined. Some stakeholders are missing or incorrectly identified.   | Vision is unclear or incomplete. Stakeholders are poorly identified or missing.                              |
| <b>2. Epics &amp; User Stories</b>                               | Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.                               | Epics and user stories are mostly complete with minor formatting or requirement issues.              | User stories are partially complete with limited Agile structure.                       | User stories are incomplete, unclear, or do not follow Agile practices.                                      |
| <b>3. Acceptance Criteria</b>                                    | Acceptance criteria are specific, measurable, testable, and aligned with all user stories.                                                          | Acceptance criteria are mostly clear and testable with minor gaps.                                   | Acceptance criteria are partially defined and lack clarity or completeness.             | Acceptance criteria are missing or not aligned with user stories.                                            |
| <b>4. Product Backlog Prioritization &amp; Sprint Planning</b>   | Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.                                              | Backlog prioritization and sprint planning are mostly appropriate with minor issues.                 | Prioritization or sprint planning is partially complete with limited justification.     | Backlog prioritization and sprint planning are poorly organized or incomplete.                               |
| <b>5. Story Point Estimation, Sprint Goal &amp; Presentation</b> | Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.    | Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear. | Estimation or sprint goal lacks justification. Presentation is adequate but incomplete. | Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete. |